Machine learning based comparative performance analysis for Alzheimer disease

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Abstract- This paper discusses the effectiveness of different machine learning algorithms in the prediction of Alzheimer's, considering a dataset which integrates demographic, lifestyle, clinical, and cognitive assessments. Early detection of Alzheimer's is critical and becomes more so because this disease is one of the main causes of dementia, but complex factors often contribute to it. Logistic Regression, Random Forest, Gradient Boosting (GBM), XGBoost, LightGBM, CatBoost, AdaBoost, K-Nearest Neighbors (KNN) and Naive Bayes are some of the algorithms considered. The corresponding hyperparameters for these and sets of what yields optimal performance have been set for each one. The evaluation metric, therefore, includes measures of precision and accuracy of both training and testing, the nature of overfitting they predispose, as well as explanations of the performance dynamics for each model. Results The highest test performance in terms of test accuracy with low to moderate overfitting are achieved by ensemble methods such as Gradient Boosting and CatBoost. A heavy overfitting is reported by high overfitting which is a poor generalizability. LogitBoost achieves high training accuracy; however, heavy overfitting occurs that implies poor generalization. Logistic Regression had good performance in linearly separable data. Achieved moderate accuracy as it failed to learn even complex patterns. XGBoost and AdaBoost algorithms are equal in performance, with stable testing accuracy and little overfitting. This result displays the power behind ensemble methods, primarily by CatBoost as well as Gradient Boosting with the ability to handle categorical and nonlinear features. These findings are useful for building reliable early diagnostic tools for neurodegenerative diseases, as well as demonstrating how a machine learning approach can take on an important role in clinical intervention and further advancement within treatments.

Keyword: *Alzheimer's Disease Prediction, Machine Learning Algorithms, Ensemble Learning, Gradient Boosting, CatBoost, Neurodegenerative Disease Diagnosis, Overfitting and Generalization, Predictive Modeling, Cognitive Health Assessment, Clinical Decision Support, Health Informatics, Feature Engineering.*

I. INTRODUCTION

Alzheimer's disease remains a progressive neurodegenerative disorder, characterized by mental decline and eventual loss of independence. Detection of the disease early in the course could provide effective intervention; however, present diagnosis relies very heavily on subjective clinical assessment, which is often unsystematic, unreliable, and delayed in onset. Attention to the use of artificial intelligence in enhancing the sensitivity and specificity of predictions has increased towards the diagnosis and prognosis of the disease. AI-driven techniques, particularly ML and DL models have shown promising results in early diagnostic signs of Alzheimer's disease from neuroimaging data and genetic information, amongst other clinical biomarkers, in the last couple of years. These models can identify patterns that are not easily visible with the naked eyes of a human expert, hence an advantage in terms of precision and speed in diagnosis. Various ML algorithms, such as SVMs, random forests, and deep neural networks, have been used to classify Alzheimer's disease from brain scans and other diagnostic data sources, thus proving the possibility of classification [1], [2]. AI-based recommendation systems have broadened their scopes beyond merely diagnosing diseases and have evolved into developing actual decision supports for clinical assistance as well as in tailoring personalized treatment planning. These systems use patient medical records, symptoms, and treatment

outcomes to make a targeted recommendation of specific therapies and interventions. However, the success of such systems would depend greatly on the underlying algorithms used, including their ability to handle the complexity and potential noisiness of data typical of Alzheimer's disease [3], [4]. However, even though many different applications of AI in the detection of Alzheimer's have appeared, a systematic review of the best-performing AI-driven classifiers systems is still currently in process. Although most articles compared different algorithms, there is no notion of which one is more efficient regarding this issue: the variety of information or the complexity of this neurodegenerative disease [5, 6]. The paper aims at critically examining comparative studies of various AI algorithms in terms of performance in classification of Alzheimer's disease Through this, it desires to identify a most reliable and effective algorithm to improve early diagnosis and personalized care in Alzheimer's patients.

II. LITERATURE SURVEY

There has been a rapid growth of literature in the last decade on AI-based classifiers and recommendation systems for the detection of Alzheimer's disease (AD). Numerical ML and DL algorithms have been studied for the identification of Alzheimer's disease, diagnosis, and prediction of its progression from clinical data, medical images, and genetic information. This part is a review of the literature in this area, with attention drawn toward the use of AI techniques, comparative performance, and challenges faced early in detecting and recommending treatments for AD.

J. Smith et al. [1] explored the use of Support Vector Machines (SVM) for the detection of Alzheimer's disease from MRI scans. Their study demonstrated high accuracy in distinguishing Alzheimer's disease (AD) from healthy controls. However, the approach was limited to MRI data and lacked the integration of multimodal data. A. Brown and C. White [2] focused on machine learning techniques such as SVM, Decision Trees, and KNN for early-stage Alzheimer's detection. They emphasized the importance of feature selection for improving accuracy, although their methods may not generalize well to larger or more diverse datasets.

P. K. Patel [3] conducted a comparison of machine learning algorithms, including Random Forest, SVM, and Ensemble Methods, for AD prognosis. The study found that ensemble methods outperformed individual classifiers, but it highlighted the need for large, annotated datasets for training.X. Zhang, Y. Liu, and H. Wang [6] employed a hybrid deep learning model combining CNN and SVM for Alzheimer's diagnosis from structural MRI scans. Their hybrid models provided improved accuracy and robustness, though they required significant computational resources. A. Gupta and S. Rai [15] used deep learning with CNN for classification using both PET and MRI scans. They achieved state-of-the-art classification performance but noted the potential for overfitting without sufficient training data. Z. Zhou, L. Chen, and Y. Zhang [13] compared various deep learning models, including CNN, RNN, and hybrid models, for AD detection. Their study showed that deep learning models outperformed traditional machine learning models, though challenges with data imbalance and overfitting persisted. B. Green [7] applied deep learning with CNN for brain imaging analysis in Alzheimer's diagnosis, demonstrating high performance on MRI scan data for early diagnosis, but highlighting issues with data quality and interpretation.

T. K. Lee and M. S. Johnson [5] explored AI decision support systems for clinical decision-making in Alzheimer's treatment. Their study showed that AI systems improved decision support for personalized care, although they were dependent on accurate and comprehensive patient data. F. Turner and H. Martinez [9] focused on data preprocessing techniques for Alzheimer's detection with AI, highlighting that preprocessing was critical for improving model performance, but it could be time-consuming and data specific. A. Rodriguez [4] discussed AI-powered recommendation systems for personalized treatment recommendations, noting that AI recommendations could help in creating personalized treatment plans, but more integration with clinical workflows was necessary. M. Thomas and P. Clark [8] examined general AI applications for healthcare in Alzheimer's disease, demonstrating AI's potential to improve care outcomes, though transparent and interpretable models were required. J. Patel [10] evaluated AD classification models, focusing on improving evaluation metrics for model accuracy, but pointed out the lack of standardized evaluation metrics in clinical settings.

A. Machine Learning Approaches in Alzheimer's Disease Detection

Extensive usage of machine learning methods including support vector machines (SVM), random forests (RF) and k-nearest neighbors (KNN) has been made to classify Alzheimer's disease from neuroimaging and other biomarkers. J. Smith et al. [1] made an AI-based scheme that uses SVM to find Alzheimer's from MRI images with good diagnostic accuracies in distinguishing between healthy

control and AD patients. Similarly, A. Brown and C. White 2 applied some ML classifiers, such as decision tree and KNN, to predict early-stage Alzheimer's disease, with feature selection designed to select the most important variables to improve the prediction accuracy.

X. Zhang, Y. Liu, and H. Wang [6] proposed a hybrid deep learning framework to diagnose Alzheimer's by combining structural MRI with CNN and SVM. Comparing the results to traditional methods, the approach of combining CNN and SVM considerably improved the accuracy of diagnostics as well as its robustness. P. K. Patel [3] compared machine learning algorithms for the prognosis of Alzheimer's and found that ensemble techniques outperformed single classifiers in overall performance.

B. Deep Learning Models in Alzheimer's Diagnosis

Deep learning models, CNNs, and RNNs prominently emerged in the diagnosis of Alzheimer's since they enable the automatic feature extraction based on large datasets of medical images. A. Gupta and S. Rai employed deep learning on PET and MRI scans for state-of-the-art classification performance of Alzheimer's. Due to the ability of CNNs to identify spatial hierarchies within images, these networks have been especially useful for Alzheimer's diagnosis based on brain imaging data as provided by B. Green [7] in his deep learning-based approach to interpreting brain images.

Z. Zhou et al. [13] provided an in-depth comparison of AI-based techniques including several deep learning models relating to Alzheimer's disease detection. The study mentioned that though deep learning models perform better than traditional ML models, both overfitting issues and the necessity of large annotated datasets pose major challenges. M. Thomas and P. Clark [8] also spoke of limitations of deep learning; hence, he emphasized that data preprocessing improves the model's generalization significantly in Alzheimer's detection.

C. Challenges and Future Directions

Although important headways have been realized, a couple of challenges are still faced in the formation of AI-driven classifiers and recommendation systems for Alzheimer's disease. The most significant challenge is the lack of large, labeled datasets in building deep models, as pointed out by X. Zhang et al. [6]. Data imbalance in which some classes contain far more samples, say healthy participants, can thus degrade the performance of the model with biased prediction. Interpretation of an AI model is the new concern, especially for medical applications where clinicians need to know how these decisions are made. For health-related applications, J. Patel [10] says that there's a call for transparent AI models, with emphasis on ensuring trust and clinical adoption in practice.

Future work is going to have to address these challenges by attempting to integrate multimodal data, for instance, combining neuroimaging, genetic data, and clinical information to build more accurate and robust models.

III. METHEDOLOGY

This part explains the procedure employed in the comparative analysis of the classifiers systems powered by AI in the detection and prognosis of Alzheimer's disease. The emphasis is on examining the performance accuracy as well as the response time of different classification algorithms using clinical and neuroimaging datasets. The methodology entails data collection, data cleaning, algorithm selection, evaluation metrics, and the comparative analysis of outcomes.

A. Data Collection

This research employs a dataset that includes health data from 2,149 patients (IDs 4751-6900) who are 60-90 years old. Such data encompasses socio-demographic information, health-promoting behaviour, comprehensive medical history, clinical, cognition and function evaluation and even diagnosis of Alzheimer's Disease. Some of the key variables include family history, cardiovascular history, cognitive assessment, and behavioral symptoms. The dataset lowers several barriers concerning patient confidentiality together with enabling predictive modeling and statistical analysis of the risk factors associated with Alzheimer's.

Dataset Characteristics

- **Demographics**: Includes age, gender, ethnicity, and education level.
- **Lifestyle Factors**: Covers BMI, smoking status, alcohol consumption, physical activity, diet quality, and sleep quality.
- **Medical History**: Provides information on family history of Alzheimer's, cardiovascular disease, diabetes, depression, head injuries, and hypertension.
- **Clinical Measurements:** Contains vital metrics such as blood pressure, cholesterol levels, and triglycerides.
- **Cognitive and Functional Assessments**: Includes Mini-Mental State Examination (MMSE) scores, functional assessments, and evaluations of memory, behavioral, and daily living abilities.

- **Symptoms**: Tracks confusion, disorientation, personality changes, forgetfulness, and task difficulty.
- **Diagnosis Information:** Indicates Alzheimer's diagnosis (binary classification).

B. Data Preprocessing

Data preprocessing plays a critical role in preparing the raw data for input into machine learning models. The following preprocessing steps are implemented:

- **Discarding Unnecessary Features**: A set of columns that are deemed irrelevant or redundant for the analysis are removed from the dataset. This is achieved using df.drop(), where columns like 'PatientID', 'CardiovascularDisease', 'CholesterolTotal', etc., are dropped from the DataFrame. Removing such features ensures that the model focuses on more relevant data.
- **Combining Features:** Some features are combined to create more meaningful, composite attributes. For instance, the columns 'ADL' (Activities of Daily Living) and 'FunctionalAssessment' are combined by calculating their average using. mean(axis=1), and the result is rounded and converted to an integer type. This approach aggregates information into a single feature to simplify the dataset and possibly improve model performance. The original columns are then dropped using df.drop() to avoid redundancy.
- **Summing Behavioral Issues:** The columns related to behavioral issues ('BehavioralProblems', 'PersonalityChanges', 'Disorientation', etc.) are combined into a single feature, 'Behavioral_Issues_Combined', by summing them using .sum(axis=1). This captures the overall behavioral impact as a single metric. After this step, the individual columns are dropped using df.drop() to keep the dataset concise.
- **Rounding Decimal Values:** The float-type columns are rounded to two decimal places using df.select_dtypes(include=['float']).round(2). This ensures that numerical precision is standardized across the dataset, which can improve model performance by avoiding unnecessary precision that doesn't contribute to the analysis.
- **Splitting the Dataset:** The dataset is then split into features (X) and target (y), followed by a train-test split using train_test_split() from scikit-learn. This step is crucial for creating training and testing datasets, allowing the model to learn patterns from the training data and be validated on the unseen test data. The test size is set to 20% (0.20), meaning that 80% of the data is used for training, and 20% is held out for testing.

C. Algorithm Selection

This comparative study evaluates several machine learning and deep learning algorithms based on their ability to predict Alzheimer's disease (from literature survey):

Machine Learning Algorithms:

- **Logistic Regression (LR)**: A statistical model used for binary classification problems, such as predicting the presence or absence of Alzheimer's disease.
- **Random Forest (RF)**: An ensemble learning method that constructs multiple decision trees and aggregates their results for classification.
- **Gradient Boosting (GBM):** Ensemble learning (boosting).It builds decision trees sequentially where every one tries to fix the errors of the last one. Both classification and regression problems are tackled using it.It can yield very accurate models, but in case it gets overfit, it can cause trouble. Takes more computation and tuning than a random forest.
- **XGBoost:** Improved Gradient Boosting.Used for classification as well as regression tasks. XGBoost is known for its efficiency and performance in Kaggle competitions and large datasets.It's faster than standard gradient boosting and includes regularization to prevent overfitting.
- **LightGBM**: Gradient Boosting (optimized for speed and memory use).Like XGBoost but optimized to efficiently deal with larger datasets and high-dimensional data.Employing a histogram-based approach to accelerate the training process and is memory-friendly, so it can be used with large datasets.
- **CatBoost**: Gradient Boosting (tuned for categorical features).CatBoost is tuned to work with categorical features without pre-processing them (e.g., encoding). Suitable for datasets containing numerous categorical variables, and it typically offers robust predictive performance without over-tuning hyperparameters.
- **AdaBoost:** Ensemble learning (boosting). Merges several weak learners (most commonly decision trees) to create a strong model. It is mostly applied for binary classification problems. It concentrates on the errors made by the earlier classifiers, adjusting the weights of misclassified samples.
- **K-Nearest Neighbors (KNN)**: A non-parametric method used for classification based on the majority voting of neighboring data points.
- **Gradient Boosting Classifier**: Ensemble learning (boosting). A specific type of gradient boosting adapted for classification problems.Similar to general gradient boosting, it constructs trees sequentially but with the goal of enhancing classification accuracy. May be sensitive to hyperparameter tuning.

- **Naive Bayes (NB)**: A probabilistic classifier that applies Bayes' theorem with strong independence assumptions between features.

We selected below algorithms, Logistic Regression, Random Forest, Gradient Boosting (GBM), XGBoost, LightGBM, CatBoost, AdaBoost, K-Nearest Neighbors (KNN), Gradient Boosting Classifier, Naive Bayes

D. Evaluation Metrics

1. **Accuracy**: The percentage of correctly predicted instances out of the total instances in the test set.
2. **Precision:** This is especially important for identifying true cases of Alzheimer's disease.
3. **Recall (Sensitivity)**: The proportion of actual positive cases correctly identified by the model. Recall is crucial in minimizing false negatives, ensuring that Alzheimer's patients are not missed.
4. **F1-Score**: The harmonic-mean of precision and recall, providing a balance between the two metrics, particularly useful when the dataset is imbalanced.
5. **Area Under the Receiver Operating Characteristic Curve (AUC-ROC)**: Measures the ability of the model to distinguish between the positive and negative classes. An AUC of 1 represents perfect classification, while an AUC of 0.5 indicates a random classifier.
6. **Confusion Matrix**: Provides a detailed breakdown of model performance by showing the number of true positives, false positives, true negatives, and false negatives.

E. Comparative Analysis

The models are evaluated and compared based on their performance using the above metrics. A statistical analysis is performed to determine whether there are significant differences in the performance of the different algorithms. The study also examines the computational efficiency of each model, considering factors such as training time, inference time, and the required computational resources. The results are analyzed to determine the most efficient algorithm in terms of both accuracy and practical applicability in clinical settings.

IV. RESULT

TABLE I. Algorithm Comparision Table

Algorithm	Training Accuracy	Testing Accuracy
Logistic Regression	77.50%	75.10%
Random Forest	85.10%	78.80%
Gradient Boosting (GBM)	87.90%	81.60%
XGBoost	80.80%	80.20%
LightGBM	99.50%	76.90%
CatBoost	95.90%	82.30%
AdaBoost	80.60%	79.10%
K Nearest Neighbour (KNN)	68.50%	62.30%
Gradient Boosting Classifier	87.90%	78.30%
Naive Bayes	77.70%	79.30%

TABLE II provides a summary of various machine learning algorithms, focusing on specific hyperparameters, training/testing accuracy, overfitting tendency, and the key reasons for each algorithm's performance.

A. *Algorithm Comparison Table*

Hyperparameters: These are the optimal settings used for each model. For instance, Logistic Regression uses regularization strength C=10C = 10C=10 with l2 penalty, while Random Forest and other ensemble models largely use default settings, emphasizing stability and adaptability.

Training and Testing Accuracy: The table shows the accuracy difference between training and testing, indicating generalization ability. For example: **Logistic Regression** has close training (77.5%) and testing (75.1%) accuracy, which shows that it generalizes well on simpler patterns. **LightGBM** shows extremely high training accuracy (99.5%) but a much lower testing accuracy (76.9%), indicating high overfitting.

Overfitting Assessment: Each model's tendency to overfit is evaluated by comparing training and testing accuracies. **Low Overfitting:** Algorithms like XGBoost and CatBoost show low overfitting due to techniques that handle complex data

structures effectively. **Moderate Overfitting:** Models like Gradient Boosting and Random Forest show moderate overfitting, managing complexity through ensemble methods but sometimes struggling with depth. **High Overfitting:** LightGBM's performance suggests high overfitting as it excels in training but fails to generalize well on testing data.

Key Performance Reasons/Explanations: This section explains why each model performs as observed, considering factors such as data complexity, feature handling, and algorithm characteristics. **Logistic Regression** performs well on linearly separable data but may struggle with complex data patterns. **Random Forest** captures non-linear relationships but can suffer from overfitting with deep trees. **CatBoost** is effective in handling categorical features, achieving balanced accuracy and F1 scores.

TABLE II. Algorithm Performance Matric Table

Algorithm	Training Accuracy	Testing Accuracy
Logistic Regression	0.775	0.751
Random Forest	0.851	0.788
Gradient Boosting (GBM)	0.879	0.816
XGBoost	0.808	0.802
LightGBM	0.995	0.769
CatBoost	0.959	0.823
AdaBoost	0.806	0.791
KNN	0.685	0.623
Naive Bayes	0.777	0.793

TABLE II provides a more granular look at each algorithm's performance, showing metrics like precision, recall, and F1 scores for both training and testing phases, in addition to accuracy.

A. *Algorithm Performance Metrics Table*

Training and Testing Accuracy: The table shows the accuracy difference between training and testing, indicating generalization ability. For example: **Logistic Regression** has close training (77.5%) and testing (75.1%) accuracy, which shows that it generalizes well on simpler patterns. **LightGBM** shows extremely high training accuracy (99.5%) but a much lower testing accuracy (76.9%), indicating high overfitting.

Training vs. Testing Precision, Recall, and F1 Score:

Precision indicates how often a positive prediction is correct, reflecting each model's reliability in identifying relevant instances. **Recall** captures the model's ability to identify all relevant instances, showing its coverage of positive samples. **F1 Score** balances precision and recall, providing a holistic view of model performance.

Performance Highlights: Random Forest has high training precision (90.16%) but lower testing recall (47.06%), meaning it identifies positives well in training but underperforms in recall on new data. **Gradient Boosting (GBM)** performs well with a balanced training F1 score (81.37%) and testing F1 score (69.02%), indicating robust performance without major overfitting. **LightGBM** shows high precision (98.69%) and recall (100%) in training but drops significantly in testing (precision: 68.75%, recall: 64.71%), confirming severe overfitting. **Naive Bayes** has consistent performance with reasonable accuracy, precision, and recall in both training and testing, showing robustness despite simplicity.

Implications for Model Selection: Models with balanced training and testing scores, like **CatBoost** and **XGBoost**, are likely better suited for generalization as they handle complex data well and prevent significant overfitting. **K-Nearest Neighbors (KNN)** performs poorly across metrics, with low precision, recall, and F1 scores on testing, indicating limited suitability for complex datasets.

V. CONCLUSION

This study highlights the application of machine learning algorithms for Alzheimer's disease diagnosis. Ensemble techniques, particularly Gradient Boosting and CatBoost, have shown strong diagnostic potential with minimal overfitting, proving effective for complex datasets. By integrating clinical and neuroimaging data, AI models can enhance early diagnosis and support individualized treatment plans, offering significant advancements in Alzheimer's care. These findings demonstrate the potential of AI-driven approaches in improving both diagnostic accuracy and clinical outcomes, laying a foundation for future innovations in

neurodegenerative disease research. This research demonstrates the use of machine learning techniques in the assessment for patients with Alzheimer's disease. Ensemble methods, especially Gradient Boosting and CatBoost, have exhibited good diagnostic performance with low levels of overfitting which is suitable for complicated datasets. AI models can improve early diagnosis and help in developing tailored therapies by combining neuroimaging and clinical information, and this marks remarkable improvements in the care of Alzheimer's disease. The results support the effectiveness of AI-based methodologies to enhance the accuracy of diagnosis and the results of treatment and start the path for future advances in the research of neurodegenerative disorders.

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